



SOFTWARE REQUIREMENTS SPECIFICATION

Guesthouse ReservationPlatform

A Full-Stack Web Application with Role-Based Property Operations Management Prepared for
the Ethiopian Guesthouse and Small-Property Accommodation Market

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1. Introduction

1.1 Purpose of this Document

This Software Requirements Specification (SRS) defines the functional and non-functional requirements for the Guesthouse Reservation Platform. It describes the system's scope, the staff and customer roles that operate it, the workflows each role follows, the data model, the technology stack, and the quality and security standards the platform must meet. This document supersedes earlier drafts that modeled the platform around only three generic roles (Guest, Owner, Admin) and replaces that model with a role structure that reflects how a real guesthouse is staffed and run day to day.

1.2 Project Overview

The Guesthouse Reservation Platform is a modern, full-stack web application designed to streamline the process of searching, booking, and managing guesthouse reservations for the Ethiopian hospitality market. It connects travelers seeking accommodation with guesthouse businesses, and it gives each guesthouse's staff — not just its owner — dedicated tools to run daily operations: taking reservations, checking guests in and out, collecting payments, and preparing rooms.

In Ethiopia, many guesthouses still rely on manual, phone-based reservation handling, which leads to missed bookings, double bookings, and poor visibility for travelers. Small and mid-sized guesthouses typically lack dedicated software to manage reservations, front-desk operations, payments, and housekeeping in an organized, auditable way. This platform is built specifically for that market, with workflows that mirror real guesthouse operations, and with future support for local payment methods such as Telebirr and Chapa.

1.3 Problem Statement

Traditional guesthouse reservation systems suffer from a combination of technical and operational gaps:

- Customers cannot easily compare guesthouses, prices, or availability online.
- Room availability is not updated in real time, causing double bookings.
- Owners and staff spend significant time managing reservations manually by phone or paper log.
- There is no single system of record connecting the front desk, the cash office, and the supervisor — so a room can be marked "available" for booking while it is still dirty or awaiting maintenance.
- Cash handling and payments are not tracked consistently, making daily reconciliation and financial reporting difficult.
- Owners have no reliable, centralized way to monitor staff activity, occupancy, or revenue across their property.

The platform addresses these gaps by giving each functional area of guesthouse operations — reservations, cashiering, supervision, and administration — its own accountable role within a single connected system.

1.4 Objectives

The overarching aim of this project is to develop a secure, user-friendly, and operationally realistic web-based Guesthouse Reservation Platform that improves the efficiency of reservation management and daily property operations. Specific objectives:

- Allow guests to search, compare, and book guesthouse rooms online.
- Prevent double booking through real-time, transaction-safe availability checks.
- Give front-desk staff (Receptionist) the tools to manage walk-in and phone reservations.
- Give financial staff (Cashier/Accountant) the tools to record payments, issue invoices, and reconcile daily income.
- Give the Supervisor a clear, current view of room status and cleaning tasks.
- Give guesthouse owners a consolidated dashboard of bookings, staff activity, and revenue.
- Give the platform Admin the tools to onboard guesthouses, manage system-wide accounts, and maintain platform health.
- Provide secure authentication and strict role-based access control across all six roles.
- Deploy a responsive, reliable application suitable for both desktop front-desk use and mobile guest use.

2. Project Scope

2.1 In Scope

- User registration and authentication for all roles (Guest, Owner, Receptionist, Cashier/Accountant, Supervisor, Admin).
- Guest profile management, guesthouse search, filtering, and comparison.
- Real-time room availability checking and online room booking.
- Manual (walk-in / phone) reservation entry by Receptionist.
- Guest check-in and check-out processing, including ID capture at check-in.
- Reservation confirmation, modification, cancellation, and booking history.
- Payment recording, invoice/receipt generation, refunds, and daily cash reconciliation.
- Room status tracking and Supervisor task assignment and completion.
- Guest reviews and ratings, with owner responses and moderation.
- Owner dashboard covering rooms, pricing, staff, reservations, and reports.
- Admin tools: guesthouse approval, system-wide user management, reporting, backups.
- Role-based access control (RBAC) enforced on every module and API endpoint.
- Reporting and analytics scoped to each role (financial, occupancy, staff-performance, system-health).
- Secure payment processing, with future integration of Telebirr and Chapa.

2.2 Out of Scope

- Flight booking
- Hotel chain / large multi-brand hospitality management
- Restaurant reservations
- Vehicle rentals
- Travel insurance
- Tour guide booking
- Native mobile application (planned for Phase 2)
- SMS notifications (planned for Phase 2)
- AI-powered recommendations (planned for Phase 3)

3. Stakeholders and System Roles

3.1 Role Hierarchy

The platform is used by one external stakeholder (the Guest) and five internal roles that together operate a guesthouse and the platform itself. The five internal roles are organized in two tiers: a platform tier (Admin) that operates across all guesthouses on the platform, and a property tier (Owner, Receptionist, Cashier/Accountant, Supervisor) that operates within a single guesthouse.

- Platform Tier
 - Admin — oversees the entire platform across all guesthouses: approves new guesthouse listings, manages every user account, maintains system health, and produces platform-wide reports.
- Property Tier (per guesthouse)
 - Owner — top of the property's staff hierarchy. Owns the business relationship with the platform, sets pricing, approves reservations policy, and supervises the property's own staff.
 - Receptionist — reports to the Owner. Runs day-to-day front-desk operations.
 - Cashier/Accountant — reports to the Owner. Runs day-to-day financial operations.
 - Supervisor — reports to the Owner (and coordinates daily with Receptionist). Runs room-readiness operations.

A booking made by a Guest flows through Receptionist for confirmation, through the Cashier/Accountant for payment, and through the Supervisor for room readiness — with the Owner able to see and, where necessary, override any step, and the Admin responsible for the platform infrastructure that makes all of this possible.

3.2 Role Descriptions and Responsibilities

Role	Main Responsibilities	Reports To
Guest (Customer)	Registers/logs in, searches and compares guesthouses, books and pays for rooms, views booking history, writes reviews, manages own profile.	N/A (external user)
Owner	Manages the business: sets room prices and availability policy, approves/rejects reservations and refund exceptions, supervises and creates accounts for property staff, reviews occupancy and financial reports, handles escalated customer issues, responds to reviews. Receives booking requests, checks room availability, confirms	Platform (self-managed business)
Receptionist	reservations, enters walk-in/phone bookings, registers guests, performs check-in and check-out, answers customer questions, hands off checked-out rooms to Supervisor. Collects payments, records deposits, prepares invoices and receipts, processes refunds, manages daily income and expenses, prepares financial	Owner
Cashier/Accountant	reports and end-of-day cash reconciliation. Views assigned rooms and cleaning tasks, updates room status (dirty / cleaning / clean / inspected / maintenance), reports maintenance issues, confirms room readiness before it can be sold again.	Owner
Supervisor	Manages user accounts across the platform, approves new guesthouse listings, maintains and monitors the reservation system, performs backups, resolves technical issues, generates system-wide reports.	Owner
Admin		Platform (technical operator)

4. User Journeys and Workflows

4.1 Guest Booking Workflow

- 1.User lands on the homepage and views featured guesthouses and the search bar.
- 2.User searches by location/city, selects check-in and check-out dates, and applies filters (price, amenities, rating).
- 3.Platform displays available guesthouses with name, location, rating, price, and number of available rooms.
- 4.User opens a guesthouse listing to view description, images, amenities, reviews, room types, and pricing.
- 5.User selects a room, confirms dates and guest count, and views the total price calculation.
- 6.User enters personal information, selects a payment method, reviews the booking summary, and accepts terms and conditions.
- 7.User completes payment, receives a confirmation email, and is issued a booking reference number.
- 8.User can later view booking history, cancel or modify a booking (subject to the cancellation policy), and write a review after the stay is complete.

4.2 Receptionist Front-Desk Workflow

- 1.Views the daily arrivals and departures list and the live room-availability calendar.
- 2.Receives phone or walk-in booking requests and creates a manual reservation after confirming availability.
- 3.Confirms, modifies, or cancels reservations on a guest's behalf when requested.
- 4.At arrival, registers the guest's identification details and performs check-in, which marks the room as occupied.
- 5.Coordinates with the Cashier/Accountant to confirm payment status before finalizing check-in.
- 6.At departure, performs check-out, which marks the room as vacant-dirty and automatically creates a Supervisor task.
- 7.Answers guest questions and logs customer requests or complaints for the Owner.

4.3 Cashier / Accountant Payment Workflow

- 1.Receives payment details from a reservation (online payment) or from Receptionist (walk-in payment).
- 2.Records the payment method (cash, card, or future mobile-money integration), amount, and reservation reference.
3. Generates and issues an invoice or receipt to the guest.
- 4.Processes refunds for eligible cancellations according to the refund policy, subject to Owner approval for exceptions.
- 5.Reconciles daily cash and digital receipts against the system's reservation and payment records.
- 6.Produces daily, weekly, and monthly financial reports for the Owner.

4.4 Supervisor Workflow

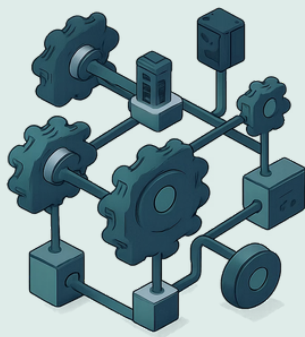
- 1.Views a task list of rooms requiring cleaning, generated automatically at check-out or manually by Receptionist/Owner.
- 2.Updates room status through the cleaning lifecycle: dirty → cleaning in progress → clean → inspected.
- 3.Flags a room as "maintenance required" if a defect is found, removing it from bookable inventory until resolved.
- 4.A room only becomes bookable again in the reservation system once the Supervisor marks it clean and inspected.

4.5 Owner Property-Management Workflow

- 1.Registers the guesthouse and submits it for platform approval.
- 2.Uploads guesthouse images, descriptions, room types, and seasonal pricing.
- 3.Creates and manages staff accounts (Receptionist, Cashier/Accountant, Supervisor) and their permissions.
- 4.Reviews incoming reservations and approves or overrides them when needed.
- 5.Monitors occupancy, revenue, and staff activity through role-specific dashboards.
- 6.Responds to guest reviews and manages escalated customer issues.

4.6 Admin Platform-Oversight Workflow

- 1.Reviews pending guesthouse listings and approves or rejects them.
- 2.Creates, suspends, or deletes user accounts across all guesthouses, including resetting the Owner account for a property.
- 3.Monitors platform-wide activity, system logs, and performance.
- 4.Performs or verifies scheduled backups and disaster-recovery readiness.
- 5.Generates platform-wide reports and monitors for fraudulent or abusive activity.



5. Business Rules

5.1 Reservation Rules

- A room cannot be booked twice for overlapping dates.
- Check-out date must be after check-in date; minimum stay duration is 1 night.
- Guests can only book, and staff can only sell, rooms currently marked available and clean.
- A reservation must be confirmed by the system (or by Receptionist for manual bookings) before payment is finalized.
- Guests can cancel up to 24 hours before check-in for a full refund; later cancellations follow the property's refund policy.
- Only Receptionist, the Owner, or the automated online-booking flow may create a reservation; check-in may only be performed by Receptionist or the Owner.

5.2 Guesthouse and Staffing Rules

- Only approved guesthouses appear in public search results.
- Only a verified Owner can register a guesthouse and add rooms; the guesthouse must have at least one room to be listed.
- Only the Owner (or the Admin, for support purposes) can create Receptionist, Cashier/Accountant, or Supervisor accounts for that property.
- Staff accounts (Receptionist, Cashier/Accountant, Supervisor) are scoped to a single guesthouse and cannot access another property's data.
- Owners can update room availability and pricing, including seasonal pricing, in real time.

5.3 Payment and Financial Rules

- Payment must be completed, or explicitly marked "pay at property," before a reservation is confirmed.
- Only the Cashier/Accountant, Owner, or Admin may issue a refund; all refunds are logged with the issuing user's identity.
- Refunds are processed within 3–5 business days.
- Failed online payments may be retried up to 3 times before the reservation is released back to availability.
- Every payment must be linked to exactly one reservation and one recording staff member for audit purposes.
- Future support for Telebirr and Chapa payments will follow the same recording and reconciliation rules as other payment methods.

5.4 Supervisor / Room–Readiness Rules

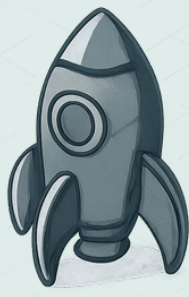
- A room automatically enters "dirty" status upon guest check-out.
- A room cannot be marked available for new bookings until the Supervisor marks it "clean" and, where required, "inspected."
- Any staff member may flag a room for maintenance; only the Supervisor or the Owner may clear a maintenance flag.

5.5 Review Rules

- Guests can only review a guesthouse after completing a stay.
- Each guest may leave one review per guesthouse, rated 1–5 stars.
- Reviews are moderated before publication; Owners may respond to published reviews.

5.6 Administrative Rules

- Only the Admin can approve new guesthouses onto the platform.
- Only the Admin can suspend or delete accounts for cause (e.g., fraud) across the platform.
- The Admin can view platform-wide operational data but does not alter a specific guesthouse's day-to-day pricing, reservations, or financial records — those remain under the Owner's authority.
- The Owner can override reservation, payment, or Supervisor decisions made by their own staff when necessary.



6. Functional Requirements

Functional requirements are grouped by the role(s) that primarily use them. All requirements are subject to the role-based access control rules defined in Section 3 and Section 8.

6.1 Shared / Authentication Requirements

- The system shall allow user registration, login, logout, and password reset for all roles.
- The system shall issue a signed token (JWT) on login and require it for all protected actions.
- The system shall enforce role-based authorization on every endpoint, scoped to the user's guesthouse where applicable.

6.2 Guest Requirements

- The system shall allow guests to search and filter guesthouses by location, price, and amenities.
- The system shall display real-time room availability and pricing.
- The system shall allow guests to make, view, modify, and cancel online reservations.
- The system shall allow guests to pay online and receive a confirmation and booking reference.
- The system shall allow guests to submit and read reviews.
- The system shall allow guests to manage their own profile.

6.3 Receptionist Requirements

- The system shall display a daily arrivals/departures list and a live room-availability calendar.
- The system shall allow Receptionist to create, confirm, modify, and cancel reservations, including manual walk-in/phone bookings.
- The system shall allow Receptionist to register guest identification details and perform check-in.
- The system shall allow Receptionist to perform check-out, which triggers a Supervisor task.
- The system shall prevent Receptionist from confirming a reservation for a room that is not available or not clean.

6.4 Cashier/Accountant Requirements

- The system shall allow the Cashier/Accountant to record payments against a reservation, including method and amount.
- The system shall generate an invoice/receipt for each recorded payment.
- The system shall allow the Cashier/Accountant to process refunds and record the reason and approver.
- The system shall allow the Cashier/Accountant to generate daily, weekly, and monthly financial reports.
- The system shall support an end-of-day cash reconciliation view comparing recorded payments to reservation totals.

6.5 Supervisor Requirements

- The system shall display a task list of rooms requiring cleaning or inspection, assigned automatically or manually.
- The system shall allow the Supervisor to update room status (dirty, cleaning, clean, inspected, maintenance).
- The system shall prevent a room from being booked while its status is dirty or under maintenance.
- The system shall allow the Supervisor to log maintenance issues with notes.

6.6 Owner Requirements

- The system shall allow the Owner to register a guesthouse and submit it for approval.
- The system shall allow the Owner to manage rooms, pricing (including seasonal pricing), and images.
- The system shall allow the Owner to create and manage staff accounts (Receptionist, Cashier/Accountant, Supervisor) for their property.
- The system shall allow the Owner to view and, where necessary, override reservations, payments, and room status set by staff.
- The system shall provide the Owner with occupancy, revenue, and staff-activity reports.
- The system shall allow the Owner to respond to guest reviews.

6.7 Admin Requirements

- The system shall allow the Admin to review, approve, or reject new guesthouse listings.
- The system shall allow the Admin to manage (create, suspend, delete) user accounts across all guesthouses.
- The system shall allow the Admin to monitor platform-wide activity and system health.
- The system shall allow the Admin to trigger or verify backups and generate platform-wide reports.
- The system shall support role-based authorization so the Admin's platform-level access does not expose one guesthouse's financial detail to another guesthouse's staff.

7. Non-Functional Requirements

Non-functional requirements define how the system performs its functions, rather than the functions themselves. They are foundational to the platform's quality, performance, and trustworthiness for both customer-facing and staff-facing use.

7.1 Performance

- Search results shall appear within two seconds under normal load.
- The system shall handle concurrent users efficiently, including simultaneous front-desk and guest activity on the same property.
- Database queries shall be optimized for speed, particularly availability checks.

7.2 Security

- Passwords shall be encrypted using bcrypt; plain-text storage is prohibited.
- JWT authentication shall be used, with session/token expiration.
- All protected routes shall require authentication and shall enforce role-based access control.
- Data shall be encrypted in transit (HTTPS/TLS) and at rest.
- Every payment, refund, and account-management action shall be logged with the acting user's identity for audit purposes.

7.3 Reliability

- Reservation transactions shall use database-level locking to prevent duplicate bookings.
- The system shall target 99.5% uptime.
- Automated daily backup and recovery procedures shall be in place.

7.4 Usability

- The interface shall be mobile-responsive for guests and clear/efficient for staff workflows on desktop devices at the front desk.
- Each role shall see a dashboard scoped to its own responsibilities, not a generic one-size-fits-all screen.
- The system shall follow accessibility best practices and support easy navigation.

8. System Architecture

The Guesthouse Reservation Platform is built on a three-tier architecture, ensuring scalability, maintainability, and a clear separation of concerns. This layered approach allows independent development, deployment, and scaling of each component.

8.1 Presentation Layer (Frontend)

- React.js user interface, with distinct views/dashboards per role.
- Responsive design using Tailwind CSS.
- Client-side routing with React Router, gated by role-based route guards.
- HTTP requests via Axios.

8.2 Application Layer (Backend)

- Node.js runtime environment.
- Express.js web framework.
- RESTful API endpoints, each protected by authentication and role-based authorization middleware.
- Business logic for reservations, payments, Supervisor task generation, and reporting.

8.3 Data Layer (Database)

- PostgreSQL relational database.
- Prisma ORM for type-safe database operations and migrations.
- Transaction support to guarantee reservation and payment consistency.

8.4 Role-Based Access Control Layer

Cutting across all three tiers, an RBAC layer maps each of the six roles to a defined set of permitted actions and data scopes (platform-wide for the Admin; property-scoped for Owner, Receptionist, Cashier/Accountant, and Supervisor; self-scoped for Guests). This is enforced both in the UI (hiding unavailable actions) and in the API (rejecting unauthorized requests).

9. Database Design

The database is structured to efficiently manage users and staff, guesthouses, rooms and their status, reservations, payments, Supervisor tasks, and reviews. The core Users table now supports six roles instead of three, and new tables capture payments and Supervisor tasks — operations that previously had no dedicated data model.

9.1 Users Table

- user_id (Primary Key)
- email, password (hashed), name, phone
- role — one of: Guest, Owner, Receptionist, Cashier, Supervisor, Admin
- guesthouse_id (Foreign Key, nullable — set for property-scoped staff; null for Guest and platform Admin)
- created_at, updated_at

9.2 Guesthouses Table

- guesthouse_id (Primary Key)
- owner_id (Foreign Key → Users)
- name, description, location, address
- rating, status (pending / approved / rejected)
- created_at, updated_at

9.3 Rooms Table

- room_id (Primary Key)
- guesthouse_id (Foreign Key)
- room_number, type, capacity
- price_per_night, availability_status (available / booked)
- housekeeping_status (clean / dirty / cleaning / inspected / maintenance)
- created_at, updated_at

9.4 Reservations Table

- reservation_id (Primary Key)
- guest_id (Foreign Key → Users)
- room_id (Foreign Key → Rooms)
- created_by (Foreign Key → Users — the Guest for online bookings, or Receptionist for manual bookings)
- check_in_date, check_out_date
- status (pending / confirmed / checked_in / checked_out / cancelled)
- total_price, checked_in_at, checked_out_at, created_at, updated_at

9.5 Payments Table

- payment_id (Primary Key)
- reservation_id (Foreign Key → Reservations)
- processed_by (Foreign Key → Users — the Cashier/Accountant, or system for online payments)
- amount, method (cash / card / online / mobile-money), type (payment / refund)
- status (pending / completed / failed / refunded)
- created_at

9.6 Supervisor Tasks Table

- task_id (Primary Key)
- room_id (Foreign Key → Rooms)
- assigned_to (Foreign Key → Users — Supervisor)
- status (pending / in_progress / completed)
- notes, created_at, completed_at

9.7 Reviews Table

- review_id (Primary Key)
- guest_id (Foreign Key → Users)
- guesthouse_id (Foreign Key → Guesthouses)
- rating, comment, created_at

10. Modules and Features

The platform is structured around interconnected modules, each aligned to one of the six system roles. This modular approach ensures clarity, scalability, and efficient development.

	10.1 Authentication Module • User registration, login, logout • Forgot password / reset password • JWT token management and role-based route protection
	10.2 Guest Module • Search and filter guesthouses • View guesthouse and room details • Book rooms and pay online • View reservation history; write and read reviews • Profile management
	10.3 Receptionist Module • Arrivals/departures dashboard and availability calendar • Manual reservation entry (walk-in / phone) • Guest registration, check-in, and check-out • Reservation modification and cancellation
	10.4 Cashier/Accountant Module • Payment recording and invoice/receipt generation • Refund processing • Daily cash reconciliation • Financial reporting (daily / weekly / monthly)
	10.5 Supervisor Module • Room-status task list • Room status updates (dirty / cleaning / clean / inspected / maintenance) • Maintenance issue logging
	10.6 Owner Module • Dashboard with occupancy, revenue, and staff-activity analytics • Guesthouse, room, and pricing management • Staff account creation and permission management • Reservation oversight and override authority • Report generation and review responses
	10.7 Admin Module • Platform dashboard with system-wide overview • User management across all guesthouses • Guesthouse approval and verification • Platform-wide reservation and activity monitoring • System reports, backups, and analytics

11. Technology Stack

Our platform is built upon a robust and modern technology stack, ensuring scalability, security, and a seamless user experience across all functionalities.

11.1 Frontend

- React.js — building dynamic, role-aware user interfaces
- Tailwind CSS — utility-first styling for rapid UI development
- React Router — client-side routing and role-based route guarding
- Axios — asynchronous HTTP requests to the backend API

11.2 Backend

- Node.js — server-side JavaScript runtime
- Express.js — REST API and route/middleware framework

11.3 Database

- PostgreSQL — relational database for reliability and data integrity
- Prisma ORM — type-safe queries and migrations

11.4 Authentication and Security

- JWT (JSON Web Tokens) — stateless authentication
- bcrypt — secure password hashing

11.5 Development and Deployment Tools

- Git and GitHub — version control and collaboration
- Postman — API testing and documentation
- VS Code — development environment
- Render — backend deployment (TLS, global CDN, DDoS protection, auto-deploy from Git)
- Netlify — frontend hosting and deployment

12. API Specification (Representative Endpoints)

All endpoints below require a valid JWT and enforce the role-based access control rules from Section 3. Endpoints marked with a role in parentheses are restricted to that role (or to the Owner and Admin, who may act on behalf of their scope).

12.1 Authentication Endpoints

- `POST /api/auth/register` — Register a new account (role assigned by context) — 201 Created
- `POST /api/auth/login` — Authenticate and receive a JWT — 200 OK
- `POST /api/auth/logout` — Invalidate the current session — 200 OK

12.2 Guesthouse and Room Endpoints

- `GET /api/guesthouses` — Search/filter guesthouses (?location=&price_min=&price_max=&rating=) — 200 OK
- `GET /api/guesthouses/:id` — Full guesthouse detail, rooms, and reviews — 200 OK
- `POST /api/guesthouses` (Owner) — Register a new guesthouse for approval — 201 Created
- `PUT /api/rooms/:id/status` (Supervisor) — Update a room's housekeeping status — 200 OK

12.3 Reservation Endpoints

- `POST /api/reservations` — Create an online reservation (Guest) — 201 Created
- `POST /api/reservations/manual` (Receptionist) — Create a walk-in/phone reservation — 201 Created
- `GET /api/reservations/:id` — Retrieve full reservation detail — 200 OK
- `PUT /api/reservations/:id/checkin` (Receptionist) — Perform guest check-in — 200 OK
- `PUT /api/reservations/:id/checkout` (Receptionist) — Perform guest check-out; creates Supervisor task — 200 OK
- `PUT /api/reservations/:id/cancel` — Cancel a reservation — 200 OK

12.4 Payment Endpoints (Cashier/Accountant / System)

- `POST /api/payments` — Record a payment against a reservation — 201 Created
- `POST /api/payments/:id/refund` (Cashier/Accountant) — Process a refund — 200 OK
- `GET /api/payments/reports` (Cashier/Owner) — Daily/weekly/monthly financial reports — 200OK

12.5 Supervisor Endpoints

- `POST /api/supervisor/tasks` — Create a Supervisor task for a room — 201 Created
- `PUT /api/supervisor/tasks/:id` — Update task status (in progress / completed) — 200 OK

12.6 Review Endpoints

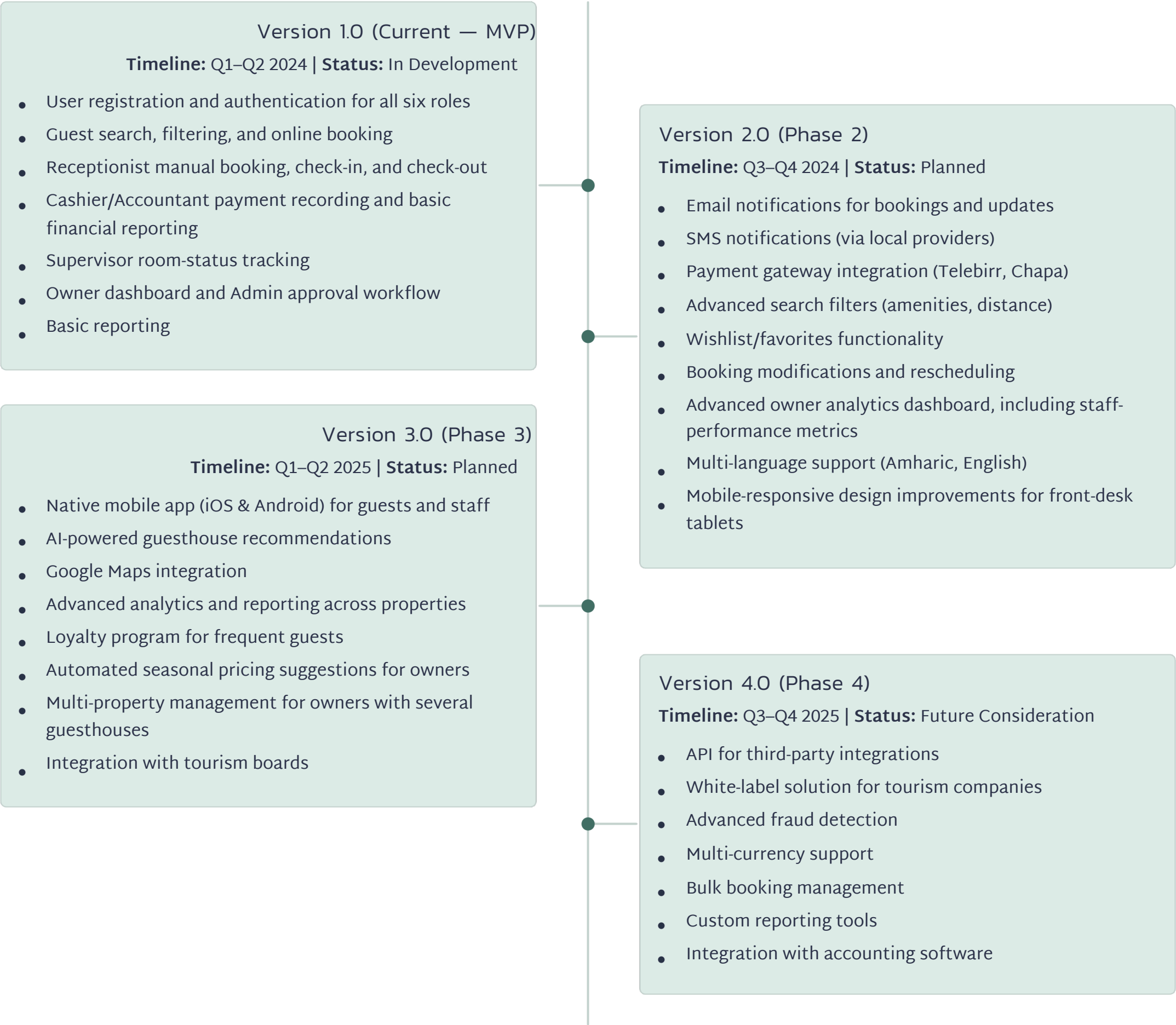
- `POST /api/reviews` — Submit a review (Guest, post-stay only) — 201 Created
- `GET /api/guesthouses/:id/reviews` — List approved reviews for a guesthouse — 200 OK

13. Risk Analysis

Risk	Impact	Mitigation Strategy
Server downtime	Users and staff cannot access the platform; lost bookings	Daily automated backups, redundant servers, 99.5% uptime SLA, monitoring alerts
Double booking	Customer dissatisfaction, revenue loss, reputation damage	Database transaction locks, date validation, real-time availability updates, automated conflict detection
Room sold while still dirty	Guest complaints, operational embarrassment	Supervisor status gating tied directly to room availability logic
Unreconciled cash/payments	Financial loss, disputes, inaccurate reporting	Mandatory payment recording per reservation, daily reconciliation view, audit logging of all Cashier/Accountant actions
Weak passwords	Account compromise, unauthorized access, data breach	Password policy enforcement, bcrypt hashing, password strength meter, optional 2FA
Unauthorized access / privilege escalation	Data theft, privacy violation, cross-property data exposure	JWT authentication, strict role-based access control, protected endpoints, regular security audits
Payment fraud	Financial loss, legal issues, customer trust loss	PCI DSS-aligned handling, SSL/TLS encryption, fraud detection, payment gateway validation
Data loss	Loss of critical business and guest data	Daily backups, disaster recovery plan, database replication, version control
SQL injection	Database compromise, data theft	Parameterized queries, input validation, ORM usage (Prisma), security testing
DDoS attacks	Service unavailability, reputation damage	Rate limiting, CDN protection, firewall rules, traffic monitoring
Scalability issues	Slow performance during peak times, poor user experience	Load balancing, database optimization, caching, horizontal scaling
Third-party service failure	Payment processing failure, service interruption	Fallback payment methods, service monitoring, vendor SLA agreements

14. Future Roadmap

Our development plan is structured in distinct phases, ensuring continuous improvement and expansion of features to meet evolving user needs and market demands.



15. Expected Outcomes

Our platform is designed to deliver significant improvements across various aspects of guesthouse management and guest satisfaction:

24/7 Online Availability

Guests can browse and book at any time, not only during phone hours.

Fewer Errors, Less Overhead

Automating reservation, payment, and Supervisor hand-offs reduces manual errors and administrative burden.

No Double Bookings

Automated, transaction-safe availability management protects inventory accuracy.

Cleaner Rooms, Fewer Complaints

Tying room availability to Supervisor status prevents selling a room that isn't ready.

Reliable Cash Control

Structured payment recording and daily reconciliation reduce financial discrepancies.

Better Customer Experience

A seamless, intuitive flow from search to confirmation to stay.

Operational Efficiency for Owners

Centralized staff and task management replaces fragmented, informal coordination.

Data-Driven Decisions

Accurate reporting and analytics let owners optimize pricing, staffing, and marketing.

Scalable, Secure Foundation

A robust technology stack capable of handling growth in bookings and properties.

Competitive Advantage

A superior digital booking and property-management experience versus manual competitors.

16. Conclusion

The Guesthouse Reservation Platform represents a significant advancement in digital accommodation booking for the Ethiopian market, addressing not only the customer-facing booking experience but also the operational reality of running a guesthouse. By replacing a generic three-role model with a role structure that reflects how guesthouses are actually staffed — Owner, Receptionist, Cashier/Accountant, Supervisor, and Admin — the platform gives every part of the business a dedicated, accountable set of tools connected through one system of record.

This document is intended to serve as the authoritative requirements baseline for design and development: it defines who uses the system, what each role is responsible for, how information flows between them, what data the system must store, and what quality and security bar the implementation must meet. Future enhancements — a dedicated mobile application, multi-gateway payment integration, AI-powered recommendations, multi-language support, and deeper analytics — should be built on top of this role-based foundation rather than around it.